

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

## 5<sup>th</sup> Semester of B. Pharm. Examination (Regular)

University Theory Examination November 2016

### PH 318 Biopharmaceutics and Pharmacokinetics

Date: 30/11/2016

Time: 10.00 a.m to 1.00 p.m

Maximum Marks: 80

#### Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section I are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

#### SECTION - I

**Q 1 Attempt all questions. Each question is of one mark.**

**20**

1. One of the following is absent in IV administration  
[A] Metabolism and excretion phase  
[B] Disposition phase  
[C] Absorption phase  
[D] None of the above
2. The process of movement of drug from its site of administration into systemic circulation is called  
[A] Disposition  
[B] Diffusion  
[C] Absorption  
[D] Drug movement
3. How do you measure bioavailability  
[A] Pharmacokinetic method  
[B] Pharmacodynamic method  
[C] A and B  
[D] Physiological techniques.
4. In compartment modelling movement of drug from compartment A to B is called  
[A] Diffusion  
[B] Dilution  
[C] Distribution  
[D] Disintegration
5. Vitamin B12 is an oily drug it is absorbed through following process  
[A] Active transport  
[B] Pore transport

- [C]Symport  
[D]Endocytosis
6. The area where carrier system is more denser for transportation  
[A] Pre metabolic site  
[B] GIT  
[C] Metabolic site  
[D]Absorption window
7. Sodium glucose transporter is  
[A] ABC transporter  
[B] Cascade transporter  
[C]Antiport  
[D]Symport
8. Thiopental is a weak acidic drug  
[A] It bypasses distribution in CSF  
[B] It doesnot distribute in CSF  
[C] Thiopental distributes in CSF to lesser extent  
[D] Thiopental distributes in CSF to larger extent
9. The brain capillaries consist of endothelial cells which are joined to one another by continuous tight intracellular junctions comprising of what is called as  
[A] BBB  
[B] astrocytes  
[C] pericytes  
[D] glial cell
10. \_\_\_\_\_ is used to promote crossing through BBB.  
[A] DMSO  
[B]Caffeine  
[C] Thiopental  
[D] Sodium salts
11. Teratogens can cross BPB in which stage?  
[A] Complete stages of Pregnancy  
[B] First trimester  
[C] Second trimester  
[D] Third trimester
12. Which of the following is a prodrug  
[A] Aspirin  
[B] Paracetamol  
[C] Dopamine  
[D] Salicylic acid
13. Who are slow acetylators among the following races  
[A] American Indians  
[B] Eskimos  
[C]Latin Americans  
[D] White People
14. BDDCS stands for  
[A] Biopharmaceutical drug dissotiation classification system  
[B] Biopharmaceutical Drug Disposition Classification System

- [C] Biological drug distribution classification system  
[D] None of the Above
15. A hypothesis that employs mathematical relationship to describe drug concentration is called  
[A] Mathematical model  
[B] Pharmacokinetic model  
[C] Both A and B  
[D] None of the above
16. Empirical models are also called  
[A] Physiological models  
[B] Compartment Models  
[C] Realistic models  
[D] Unrealistic models
17. Clearance is a ration of  
[A]  $dX/dt / C$   
[B]  $C * dX/dt$   
[C]  $dX/dt$   
[D] None of the above
18. Half life for IV bolus administration is  
[A]  $t_{1/2} = 3.14 / t$   
[B]  $t_{1/2} = 0.693/K_e$   
[C]  $t_{1/2} = 3 * K_e$   
[D] None of the above
19. Line Weaver Burke Plot is for estimation of  
[A]  $K_m$   
[B]  $V_{max}$   
[C] Both A and B  
[D]  $T_{max}$
20. Hanes Woolf plot and Woolf Augustinsson Hofstee plot are for determination of  
[A]  $K_m$   
[B]  $V_{max}$   
[C]  $T_{max}$   
[D] Both A and B

## SECTION – II

**Q 2 Attempt any TWO of the following**

- A** Define Absolute Bioavailability and Relative Bioavailability.  
Following Data was obtained for 4 formulations of Pentoxifylline in volunteers of average weight of 50 Kgs.

| Drug Product  | Dose mg/kg | AUC mcg.hr/l |
|---------------|------------|--------------|
| IV solution   | 1.2        | 450          |
| Oral Solution | 4.0        | 822          |
| Oral Capsule  | 4.0        | 736          |

05

|                |     |      |
|----------------|-----|------|
| Oral SR tablet | 8.0 | 1040 |
|----------------|-----|------|

- I. What is absolute bioavailability of Capsule and SR tablet.
- II. What is relative bioavailability of Capsule and SR tablet as compared with Oral Solution.

**B** Give Handerson Hasselbach Equations.

Given below are 2 basic drugs.

| Properties              | Diazepam | Loratidine |
|-------------------------|----------|------------|
| Molecular weight        | 285      | 383        |
| pKa                     | 3.7      | 5.0        |
| Aqueous solubility      | Slight   | Insoluble  |
| Ko/w                    | High     | 27.0       |
| %Presystemic Metabolism | Nil      | 90.0       |

05

Determine the % ionization in stomach at following pH

- I. pH 3.5
- II. pH 5.0

**C** Give definition of volume of distribution and its equation.

The Vd of 3 drugs A= 12, B= 42, C= 400 litres. Calculate the percentage of drug present outside plasma for all three drugs.

05

**Q 3** Attempt any TWO of the following

**A** If the plasma concentration of viomycin after IV bolus administration was found to be 10.0 mcg/ml and 5.5 mcg/ml at 2 hours and 4 hours respectively, assuming one compartment kinetics calculate :

- I. Half life of the drug.
- II. Concentration of drug in plasma at time  $t=0$
- III. Vd if dose administered was 300mg.
- IV. Total systemic clearance.

05

**B** Draw comparison between compartment models and physiological models.

05

**C** Derive the equation for One compartment open model for IV infusion.

05

### SECTION – III

**Q 4** Attempt any FOUR of the following

**A** What do you mean by biopharmaceutics? Enlist the factors affecting GI absorption of drugs.

05

**B** Describe factors affecting metabolism of drug.

05

**C** Differentiate Phase I and II reactions with appropriate examples.

05

**D** Describe the influence of protein binding on drug disposition and describe Kinetics of Protein drug binding.

05

**E** "Hydrophilicity and lipophilicity property of a drug decide its absorption" Comment with justification.

05

**F** Discuss the absorption of drugs from non-oral extravascular routes.

05

- Q 5    Attempt any FOUR of the following**
- |          |                                                                                                                                    |           |
|----------|------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>A</b> | Explain Renal and Non renal routes of excretion of Drugs from Body.                                                                | <b>05</b> |
| <b>B</b> | Write full name of BCS. Classify and give its objective. Give the regulatory requirement for conduction of bio-equivalent studies. | <b>05</b> |
| <b>C</b> | Define bioavailability. Give its objective. Discuss plasma level time studies for measurement of bioavailability.                  | <b>05</b> |
| <b>D</b> | Explain in detail theories of Drug Dissolution.                                                                                    | <b>05</b> |
| <b>E</b> | Explain in detail IVIVC.                                                                                                           | <b>05</b> |
| <b>F</b> | What is Non Linear Pharmacokinetics? What are causes of Non Linearity. Derive Mechalis Menton Equation.                            | <b>05</b> |
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